

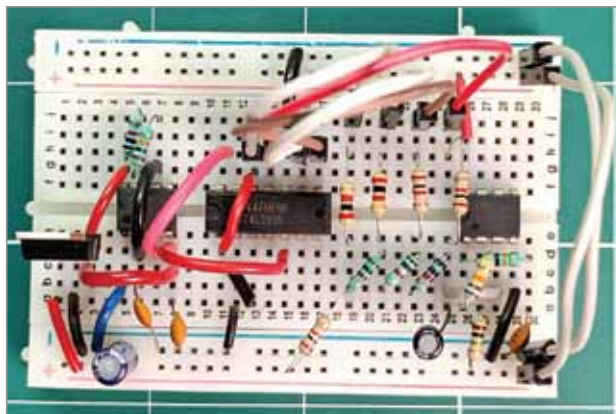


Fig. 8: Author's prototype

output is effectively buffered. Although the waveform remains similar to the smoothed case (as in Fig. 4), the op-amp improves signal stability and provides proper isolation from the load. It also prevents any loading effects on the circuit. The op-amp is powered by a 9V supply to avoid clipping and ensure adequate output swing, while an LM7805 voltage regulator (U1) delivers a stable 5V supply to the rest of the circuit.

Using the LM358 instead of only a capacitor offers several advantages. It acts as a voltage buffer, preventing loading effects and preserving signal integrity. Unlike a capacitor, which can only smooth the signal, the op-amp can also amplify it. Additionally, it enables better control over filtering through active filter configurations, provides low output impedance for efficient load driving, and improves overall signal conditioning and stability without introducing unwanted distortion.

Construction and testing

An actual-size, single-sided PCB layout for the sawtooth generator with op-amp LM358 (see Fig. 5) is shown in Fig. 6, and its component layout is shown in Fig. 7. After assembling the circuit on the PCB, mount it inside a suitable enclosure. Connect J1 on the front side of the enclosure to measure the sawtooth output, and fix the assembled PCB inside the enclosure.

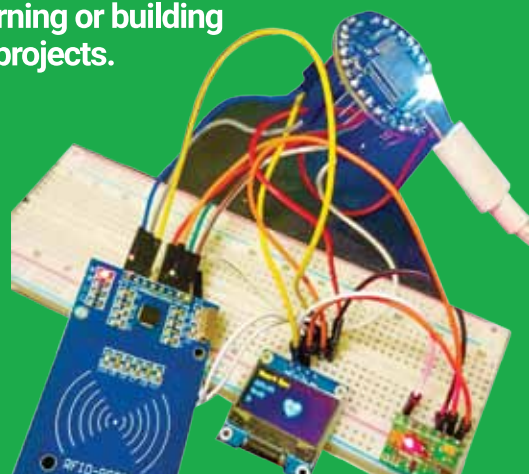
For testing, connect a 9V battery to power the circuit and observe the output on a DSO through connector J1. For user convenience, a potentiometer is not included to vary the sawtooth waveform. Based on the selected values of R1, R2, and C3, the output frequency is approximately 700Hz (see Fig. 4). However, the frequency can be adjusted by varying R1, R2, or C3, as required. **EFY**

Abhay Verma is an electronics engineer at EFY and a passionate electronics hobbyist with a keen interest in practical circuit design and DIY projects.

DIY KIT: NFC, RFID, IoT & Embedded Systems

Want to learn and master NFC and RFID technology along with an IoT-enabled embedded system?

This kit includes all the essentials to start learning or building RFID/NFC projects.



POWERED BY



IndusBoard
COIN

Super small sized development board that enables you to create solutions for many use-cases where small size is super important.



SPECIAL INTRO PRICE

₹1,999 (Inclusive of GST)

KNOW MORE / BUY NOW: efy.co.in/nfc-rfid

Developed by the Electronics For You team, for you.

DEALERS: • Om Technology Centre, Bengaluru (Mob: +919845400864)
• Bharathi Electronics: Coimbatore (Mob: +919952228618)